

Gurmeher Singh Puri

Nationality: Indian | Date of Birth: 23/06/1999 | Gender: Male

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Profile

Sixth-semester Master's student in Computer Science with a focus on **Artificial Intelligence** at the University of Freiburg, currently completing my Master's thesis at **Fraunhofer ISE** and the **Machine Learning Lab** under the supervision of **Dr. André Biedenkapp** and **Prof. Dr. Frank Hutter**. Hands-on experience across the full ML lifecycle: data curation and feature engineering, supervised regression and classification, tabular foundation models, ensemble and stacking methods, AutoML, and **Bayesian Optimisation** for hyperparameter and process tuning. Three years of prior production software-engineering experience at Samsung R&D in Python, automation, and CI/CD. Comfortable on HPC clusters, GPU workflows, and reproducible experiment design. Looking to contribute to applied ML, MLOps, or AI research roles.

Master's Thesis Research

February 2026 – August 2026

Automated Crack Detection and Process Optimisation in Laser-Assisted Splitting of SiC Wafers using Bayesian Optimisation

Fraunhofer Institute for Solar Energy Systems (ISE) · Machine Learning Lab, University of Freiburg

Supervisors: Dr. André Biedenkapp & Prof. Dr. Frank Hutter

- Designed and implemented an **end-to-end ML data pipeline** that ingests raw 1D laser-confocal profilometer scans, reconstructs the spatial axis, removes sensor dead zones, segments the signal into ~2,160 physical blocks, and cross-references each block against a Design-of-Experiments table to produce a curated 30-column tabular training dataset.
- Engineered an **unsupervised computer-vision crack-detection algorithm** (RANSAC line fitting → DBSCAN density clustering → highest-cluster surface extraction with geometric filters) to derive physically meaningful target labels from noisy sensor data.
- Implemented a **multi-objective Bayesian Optimisation loop** on top of **SMAC3** using the **ParEGO** strategy with the augmented Tchebycheff scalarisation to jointly optimise kerf loss and crack convergence over a 9-dimensional continuous configuration space.
- Developed a **stacked surrogate ensemble** combining Random Forest, XGBoost, LightGBM, CatBoost, Gaussian Processes, and tabular foundation models (**TabPFN**, **TabICL**) with a Ridge meta-learner, group-aware K-Fold cross-validation, and per-target log transforms; benchmarked against single-model baselines.
- Closed-loop **active learning** workflow that proposes new laser parameter recipes for evaluation on the physical machine.

Employment History

Working Student (ML & Image Processing), *Fraunhofer-Institut für Solare Energiesysteme ISE*

March 2024 – Present

Freiburg, Germany

- Core developer of the **Spotlob** image-processing library: automated computer-vision detection of circles, lines, and edges on high-volume scientific imagery.
- Designed modular, composable processing pipelines for feature extraction from large image datasets.
- Built supporting data infrastructure with **Apache Airflow** (workflow orchestration), **MinIO** (S3-compatible object storage), and **Flask** (webhook notifications) for reproducible experiment runs.

AI Work Student, *INATECH & Machine Learning Lab, University of Freiburg* December 2024 – July

2025
Freiburg, Germany

- Built an **AutoML** forecasting system (**AutoGluon**) for electricity-grid carbon emission intensities across Germany.
- Integrated heterogeneous large-scale data sources (ENTSO-E transparency platform, CO2map.de) covering generation forecasts, cross-border electricity trading, and carbon-intensity measurements.
- Owned feature engineering, model selection, and back-testing of the forecasting pipeline.

Software Engineer, *Samsung Research and Development Institute* August 2021 – September 2023
Noida, India

- Designed and shipped **12 internal tools and web services** in Python and JavaScript, automating workflows and saving an aggregate **>149,000 engineering hours per year**.
- Owned **CI/CD pipelines** executing smoke tests on **1,000+ physical devices** for the Samsung Knox security platform.
- Authored automated test cases in Android Studio + JUnit and managed distributed contest infrastructure on the DOMjudge system.

Software Engineer Intern, *Samsung Research and Development Institute* January 2021 – July 2021
Noida, India

- Built two internal websites with JavaScript, Python, and CSS, and contributed Samsung Knox automation test cases.

Web Developer Intern, *Bharat Heavy Electricals Limited (BHEL)* May 2019 – July 2019
India

- Developed the *Student Tracker* application using Eclipse and Tomcat 9.0.

Technical Skills

Machine Learning: Random Forest, Gradient Boosting (XGBoost, LightGBM, CatBoost), Gaussian Processes, Ridge / Lasso, stacked ensembles, AutoML (AutoGluon), Bayesian Optimisation (SMAC3, ParEGO), hyperparameter tuning, cross-validation, feature engineering

Deep & Foundation Models: TabPFN, TabICL, transformer-based tabular learners; familiar with PyTorch and Hugging Face workflows

Computer Vision & Signal Processing: OpenCV, RANSAC, DBSCAN, clustering, profilometer / image data, custom feature extraction pipelines

NLP: NLTK, spaCy (Samsung research chatbot)

Languages: Python (expert), Java, C++, SQL, Bash/Shell, JavaScript

Data & Tooling: Pandas, NumPy, SciPy, scikit-learn, Jupyter, Matplotlib; Apache Airflow, MinIO, Docker, Git, GitLab CI/CD

Scientific Computing: HPC clusters (Helix, bwUniCluster), GPU computing, SLURM, remote SSH workflows, PyCharm

Languages (Human): English (fluent), German (working proficiency)

Education

M.Sc. in Computer Science (Artificial Intelligence focus), *Albert-Ludwigs-Universität Freiburg*
October 2023 – August 2026
Currently in 6th semester; thesis defense August 2026

B.Tech. in Computer Science and Engineering, *Amity University Uttar Pradesh* July 2017 – June 2021
Noida, India

Selected Projects

Electricity Carbon Intensity Prediction

Python, AutoGluon, Pandas, ENTSO-E API

AutoML forecasting system for German grid carbon intensity; multi-source data integration and reproducible feature pipeline.

Predicting Algorithm Runtime with TabPFN

Python, TabPFN, Deep Learning

Investigated tabular foundation models against specialised deep regression baselines for algorithm runtime distribution prediction (ML Lab, University of Freiburg).

Spotlob Image-Processing Library

Python, OpenCV, Modular Pipelines

Open-source image-processing library at Fraunhofer ISE for reusable, composable feature-extraction pipelines on scientific imagery.

Chatbot for Research Assistance

Python, NLTK, spaCy

NLP-driven assistant to accelerate patent search and prior-art research at Samsung R&D.

Automation Process Tool

Python, Apache Airflow, MinIO, Docker, Flask

Internal data orchestration platform for an automated testing and validation pipeline.

Automated Data Processing Tool

Python, Pandas

Cleansing and transformation of large-scale device-performance logs at Samsung R&D.

Publications

- *High Performance Computers and Computational Science* — Dec. 2019 – Apr. 2020.
- *Cryptographic Technique for Communication Systems via the Diffie-Hellman Key Algorithm* — Dec. 2020 – Jun. 2021.
- *A Review on Cloud Computing* — Jan. 2019 – May 2019.

Achievements & Certifications

- Samsung **Employee of the Quarter** (Q1 & Q3 2022) and **Best Project Award** for the RPA SWEET Tool.
- Samsung **RPA Best Team Award**.
- Stanford Online: *Algorithms: Design and Analysis*.
- UiPath Certified Professional Automation Developer Associate.
- Microsoft Certification: Programming in HTML5 with JavaScript & CSS3.
- Google Digital Unlocked — Digital Marketing Expert.
- NSIC Entrepreneurship Orientation Program (EOP).